

P NIKHIL KUMAR NAIK

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SUMMARY

Results-driven Data Science graduate with hands-on experience in Machine Learning, Deep Learning, NLP, and Generative AI. Proficient in building end-to-end AI solutions including predictive models, NLP pipelines, and RAG-based systems using Python, TensorFlow, and PyTorch. Strong in data analysis, feature engineering, and model deployment. Seeking a Data Scientist or ML Engineer role to deliver scalable, high-impact solutions.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL
- **Machine Learning:** Supervised & Unsupervised Learning, Linear & Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, KNN, Naive Bayes, K-Means, DBSCAN, Hierarchical Clustering, PCA, Ensemble Methods (Bagging, Boosting), SMOTE, Cross-Validation, Hyperparameter Tuning
- **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), LSTM, Transfer Learning, Image Classification, Object Detection, Autoencoders, TensorFlow, PyTorch, Keras
- **Natural Language Processing (NLP) & Generative AI:** Text Preprocessing, Tokenization, Named Entity Recognition (NER), Sentiment Analysis, Text Classification, Text Summarization, Word2Vec, BERT, Transformers, LLM Integration (Groq, OpenAI API), LangChain, RAG Pipelines, Prompt Engineering, HuggingFace, Chatbot Development, AI Automation
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Power BI, Tableau, Exploratory Data Analysis (EDA), Feature Engineering
- **Frameworks & Tools:** Scikit-learn, FastAPI, Streamlit, Pinecone (Vector DB), Jupyter Notebook, VS Code, Git, GitHub
- **Databases & Cloud:** MySQL, AWS (EC2, S3 - Basic), Cloud Computing Fundamentals

PROJECTS

•AI-Powered Medical Chatbot

- Generative AI / RAG / LangChain / Groq LLM / Pinecone / HuggingFace / Streamlit
- GitHub
- Built a RAG-based medical chatbot capable of answering context-aware health queries by ingesting and indexing domain-specific medical documents into a Pinecone vector database for semantic search.
  - Integrated Groq LLM with HuggingFace embeddings and LangChain retrieval chains, reducing irrelevant response rate and improving answer relevance by approximately 35% over baseline.
  - Processed unstructured data into vector embeddings for fast semantic search and low-latency retrieval, and built an interactive conversational web interface using Streamlit for real-time user interaction.

•Music Recommendation System

- Machine Learning / Collaborative Filtering / Faiss / Scikit-learn / Python
- GitHub
- Developed a hybrid music recommendation engine combining collaborative filtering and content-based filtering to generate personalized song suggestions with 85%+ user preference match on test data.
  - Implemented Faiss for efficient approximate nearest neighbor search, reducing similarity computation time by 60% compared to brute-force cosine similarity on a dataset of 10,000+ tracks.
  - Engineered multi-dimensional song features such as genre, artist, lyrics, and audio attributes to enhance recommendation accuracy.

•Obesity Level Detection – Final Year Project

- Machine Learning / Health Analytics / LightGBM / SMOTE / Streamlit
- GitHub
- Built a multi-class ML classification system to predict 7 obesity severity levels from personal and lifestyle data, achieving 97% accuracy on the UCI Obesity Dataset.
  - Applied SMOTE oversampling to handle class imbalance, improving F1-score on minority classes by 18%; used StandardScaler and OneHotEncoder for robust feature preprocessing.
  - Trained and optimized a LightGBM classifier with cross-validation and hyperparameter tuning; deployed a user-facing prediction interface via Streamlit.

WORK EXPERIENCE

•Data Science Intern

- Strydo Technologies Pvt. Ltd.
- Tirupati, Andhra Pradesh
- Oct 2025 – Jan 2026
- Designed and deployed 5+ end-to-end ML/DL pipelines spanning classification, regression, feature engineering, model development, evaluation, and NLP tasks, reducing manual data processing time by approximately 40%.
  - Built and fine-tuned deep learning models using TensorFlow and PyTorch, achieving model accuracy improvements of 10–15% through hyperparameter tuning and ensemble techniques.
  - Developed Generative AI and NLP pipelines using transformer-based models, LangChain, and prompt engineering, enabling automated content generation and semantic text analysis.
  - Performed end-to-end data preprocessing, feature engineering, and model evaluation using **Scikit-learn, Pandas, and NumPy**, improving prediction reliability across multiple production workflows.

EDUCATION

B.Tech, Computer Science Engineering (Data Science Specialization)

2021 – 2025

Sree Vidyanikethan Engineering College, Tirupati

CGPA: 8.64 / 10.0

Intermediate – Mathematics, Physics, Chemistry (12th Grade)

2019 – 2021

Board of Intermediate Education, Andhra Pradesh

CGPA: 9.6 / 10.0

SSC (10th Grade)

2019

Board of Secondary Education, Andhra Pradesh

CGPA: 9.8 / 10.0

CERTIFICATIONS

- IBM Data Science Professional Certificate – IBM on Coursera
- Google Data Analytics Professional Certificate – Google on Coursera
- AI Foundation Certificate – Infosys Springboard
- Altair Data Science Master Virtual Internship – AICTE Certified
- MePro Level 10 English Standard Course Certification